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(i) Solving $x^2y^3 + x\left(1+y^2\right)y' = 0$

- The terms are isolated to write the equation in the standard separable form $M(x) + N(y)\frac{dy}{dx} = 0$.
- Assuming $x \neq 0$ and $y \neq 0$, the entire equation is divided by xy^3:

$$x + \left(y^{-3} + y^{-1}\right)\frac{dy}{dx} = 0$$

- Integrate both sides:

$$\int x\,dx + \int \left(y^{-3} + y^{-1}\right)dy = C$$

- Evaluating the integrals yields:

$$\frac{1}{2}x^2 - \frac{1}{2}y^{-2} + \ln|y| = C$$

- Plugging $y = 0$ into the original equation gives $0 + 0 = 0$, meaning $y = 0$ **is also a valid particular solution**.

(ii) Solving $\left(\frac{\sin y}{y} - 2e^{-x}\sin x\right) + \left(\frac{\cos y + 2e^{-x}\cos x}{y}\right)y' = 0$

- Multiply the entire equation by y (assuming $y \neq 0$):

$$(\sin y - 2ye^{-x}\sin x) + (\cos y + 2e^{-x}\cos x)y' = 0$$

- $M(x,y)$ and $N(x,y)$ are identified:

$$M(x,y) = \sin y - 2ye^{-x}\sin x$$
$$N(x,y) = \cos y + 2e^{-x}\cos x$$

- It is checked if the equation is exact by calculating the partial derivatives M_y and N_x:

$$M_y = \frac{\partial M}{\partial y} = \cos y - 2e^{-x}\sin x$$

$$N_x = \frac{\partial N}{\partial x} = -2e^{-x}\cos x - 2e^{-x}\sin x$$

- Since $M_y \neq N_x$, the equation is not exact. An integrating factor must be found.

$$K(x) = \frac{(\cos y - 2e^{-x}\sin x) - (-2e^{-x}\cos x - 2e^{-x}\sin x)}{\cos y + 2e^{-x}\cos x} = \frac{\cos y + 2e^{-x}\cos x}{\cos y + 2e^{-x}\cos x} = 1$$

- Because $K(x) = 1$ is a function of x only, the integrating factor $\mu(x)$ can be found by solving $\frac{d\mu}{dx} = \mu(x)K(x)$:

$$\mu(x) = e^{\int 1\,dx} = e^x$$

- The modified ODE is multiplied by the integrating factor e^x to make it exact:

$$(e^x\sin y - 2y\sin x) + (e^x\cos y + 2\cos x)y' = 0$$

- Now, the potential function $\Psi(x,y)$ is found such that $\frac{\partial\Psi}{\partial x} = M_{new}$ and $\frac{\partial\Psi}{\partial y} = N_{new}$. N_{new} is integrated with respect to y:

$$\Psi(x,y) = \int (e^x\cos y + 2\cos x)dy = e^x\sin y + 2y\cos x + h(x)$$

- Ψ is differentiated with respect to x and set equal to M_{new} to find $h(x)$:

$$\frac{\partial\Psi}{\partial x} = e^x\sin y - 2y\sin x + h'(x) = e^x\sin y - 2y\sin x$$

- This shows $h'(x) = 0$, so $h(x) = 0$.
- The general implicit solution is $\Psi(x,y) = c$:

$$e^x\sin y + 2y\cos x = c$$

2

Consider the first–order ODE in differential form

$$M(x,y)dx + N(x,y)dy = 0 \quad\Longleftrightarrow\quad M(x,y) + N(x,y)y' = 0.$$

An integrating factor $\mu(x,y)$ makes the equation exact, i.e.,

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N). \tag{1}$$

Expanding using the product rule gives

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x \quad \Longleftrightarrow \quad M\mu_y - N\mu_x = \mu(N_x - M_y).$$

Assuming μ depends only on the product $z = xy$, i.e., $\mu(x, y) = \mu(z)$. Then by the chain rule,

$$\mu_x = \mu'(z)y, \qquad \mu_y = \mu'(z)x.$$

Substituting into the PDE yields:

$$M(x\mu'(z)) - N(y\mu'(z)) = \mu(z)(N_x - M_y),$$

so

$$(xM - yN)\mu'(z) = \mu(z)(N_x - M_y).$$

If

$$\frac{N_x - M_y}{xM - yN} = R(xy) = R(z),$$

then an ODE in z is obtained:

$$\frac{\mu'(z)}{\mu(z)} = R(z). \tag{2}$$

Integrating gives:

$$\ln \mu(z) = \int R(z)dz + C \quad \Longrightarrow \quad \boxed{\mu(z) = C \exp\left(\int R(z)dz\right)}.$$

Hence an integrating factor of the desired form is

$$\boxed{\mu(xy) = C \exp\left(\int^{xy} R(s)ds\right)},$$

where $C \neq 0$ is an arbitrary constant (any nonzero constant multiple is also an integrating factor).

3

$$M(x, y) = 3x + \frac{6}{y}, \qquad N(x, y) = \frac{x^2}{y} + \frac{3y}{x}.$$

$$M_y = -\frac{6}{y^2}, \qquad N_x = \frac{2x}{y} - \frac{3y}{x^2}.$$

Since $M_y \neq N_x$, it is **not exact**.

If an integrating factor $\mu = \mu(z)$ with $z = xy$ is suspected, then μ satisfies

$$\mu'(z) = \mu(z)L(t, y), \quad L(t, y) = \frac{N_t - M_y}{tM - yN},$$

and if L depends only on $z = ty$, then this produces an integrating factor $\mu(ty)$.

Here (with $t = x$):

$$xM - yN = x\left(3x + \frac{6}{y}\right) - y\left(\frac{x^2}{y} + \frac{3y}{x}\right) = 2x^2 + \frac{6x}{y} - \frac{3y^2}{x},$$

and

$$N_x - M_y = \left(\frac{2x}{y} - \frac{3y}{x^2}\right) - \left(-\frac{6}{y^2}\right) = \frac{2x}{y} - \frac{3y}{x^2} + \frac{6}{y^2}.$$

Simplification gives

$$L(x, y) = \frac{N_x - M_y}{xM - yN} = \frac{1}{xy} = \frac{1}{z},$$

which indeed depends only on $z = xy$. Therefore

$$\mu'(z) = \mu(z)\frac{1}{z} \quad \Rightarrow \quad \frac{\mu'}{\mu} = \frac{1}{z} \Rightarrow \ln|\mu| = \ln|z| + C \Rightarrow \mu = Cz.$$

The simplest choice is $\boxed{\mu(x, y) = xy}$.

Multiply the original equation by xy:

$$xy\left(3x + \frac{6}{y}\right)dx + xy\left(\frac{x^2}{y} + \frac{3y}{x}\right)dy = 0$$

$$\Rightarrow (3x^2y + 6x)dx + (x^3 + 3y^2)dy = 0.$$

Now it is exact since

$$\frac{\partial}{\partial y}(3x^2y + 6x) = 3x^2 = \frac{\partial}{\partial x}(x^3 + 3y^2).$$

Find a potential Ψ with $\Psi_x = 3x^2y + 6x$:

$$\Psi(x,y) = \int (3x^2y + 6x)dx = x^3y + 3x^2 + h(y).$$

Differentiate w.r.t. y and match $\Psi_y = N_1 = x^3 + 3y^2$:

$$\Psi_y = x^3 + h'(y) = x^3 + 3y^2 \Rightarrow h'(y) = 3y^2 \Rightarrow h(y) = y^3.$$

Hence the implicit general solution is

$$\boxed{x^3y + 3x^2 + y^3 = C.}$$

Domain note: the original ODE contains $\frac{1}{x}$ and $\frac{1}{y}$, so solutions are considered on intervals where $x \neq 0$ and $y \neq 0$.

4

For the IVP $y' = ay,\ y(0) = 1$, the integral form is

$$y(t) = 1 + \int_0^t ay(s)ds,$$

and Picard iteration (successive approximations) defines

$$y_0(t) = 1, \qquad y_{n+1}(t) = 1 + \int_0^t ay_n(s)ds$$

1) Closed-form formula for $y_n(t)$

The first few iterations are computed as follows:

- $y_0(t) = 1$.
- $y_1(t) = 1 + \int_0^t a \cdot 1 ds = 1 + at$.
- $y_2(t) = 1 + \int_0^t a(1 + as)ds = 1 + at + \frac{(at)^2}{2!}$.
- $y_3(t) = 1 + \int_0^t a\left(1 + as + \frac{(as)^2}{2!}\right)ds = 1 + at + \frac{(at)^2}{2!} + \frac{(at)^3}{3!}$.

By induction, if

$$y_n(t) = \sum_{k=0}^{n} \frac{(at)^k}{k!},$$

then

$$y_{n+1}(t) = 1 + \int_0^t a \sum_{k=0}^{n} \frac{(as)^k}{k!} ds = 1 + \sum_{k=0}^{n} \frac{a^{k+1}}{k!} \cdot \frac{t^{k+1}}{k+1} = \sum_{k=0}^{n+1} \frac{(at)^k}{k!}.$$

Thus, the n -th Picard approximation is

$$\boxed{y_n(t) = \sum_{k=0}^{n} \frac{(at)^k}{k!}.}$$

2) Limiting function $y(t) = \lim_{n\to\infty} y_n(t)$

Taking $n \to \infty$,

$$\boxed{y(t) = \sum_{k=0}^{\infty} \frac{(at)^k}{k!} = e^{at}.}$$

3) Verification of the limit as a solution

Yes. Differentiating $y(t) = e^{at}$ yields:

$$y'(t) = ae^{at} = ay(t), \qquad y(0) = e^0 = 1,$$

so it satisfies the IVP.

This also matches the explicit solution formula for separable/linear equations $y' = p(t)y \Rightarrow y(t) = \kappa e^{\int p(t)dt}$. With $p(t) = a$ and $y(0) = 1$, it follows that $\kappa = 1$ and $y(t) = e^{at}$.

4) Are there other solutions?

- For the **ODE** $y' = ay$ (without an initial condition), there is the family $y \equiv 0$.
- For the **IVP** $y' = ay,\ y(0) = 1$, there are **no other solutions**: coefficients are continuous, so the existence/uniqueness theorem gives a **unique** solution on an interval (in fact globally here, since a is constant/continuous everywhere).

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(i) Exact solution

The IVP is separable:

$$\frac{dy}{dt} = y^2 \quad \Rightarrow \quad \frac{1}{y^2}, dy = dt.$$

Integrate:

$$\int y^{-2}, dy = \int 1, dt; \Rightarrow; -\frac{1}{y} = t + C.$$

Using $y(0) = 1$:

$$-\frac{1}{1} = 0 + C \Rightarrow C = -1.$$

Hence

$$-\frac{1}{y} = t - 1; \Rightarrow; y(t) = \frac{1}{1-t}.$$

This solution blows up at $t = 1$, so the (maximal) interval of existence for the IVP is $t \in (-\infty, 1)$.

(ii) First three Picard approximations y_1, y_2, y_3 and comparison

The IVP is rewritten as the integral equation

$$y(t) = 1 + \int_0^t (y(s))^2, ds,$$

and Picard iteration is applied

$$y_{n+1}(t) = 1 + \int_0^t (y_n(s))^2, ds,$$

starting from $y_0(t) = 1$.

1) First approximation

$$y_1(t) = 1 + \int_0^t (1)^2, ds = 1 + t.$$

2) Second approximation

$$y_2(t) = 1 + \int_0^t (1+s)^2 ds = 1 + \int_0^t (1 + 2s + s^2) ds = 1 + t + t^2 + \frac{t^3}{3}.$$

3) Third approximation

$$y_3(t) = 1 + \int_0^t \left(1 + s + s^2 + \frac{s^3}{3}\right)^2 ds.$$

Expanding and integrating term-by-term gives

$$y_3(t) = 1 + t + t^2 + t^3 + \frac{2}{3}t^4 + \frac{1}{3}t^5 + \frac{1}{9}t^6 + \frac{1}{63}t^7.$$

Comparison with the exact solution

The exact solution

$$y(t) = \frac{1}{1-t}$$

has the geometric-series expansion (for $|t| < 1$):

$$\frac{1}{1-t} = 1 + t + t^2 + t^3 + t^4 + t^5 + \cdots$$

- $y_1(t) = 1 + t$: matches the exact series through the t^1 term.
- $y_2(t) = 1 + t + t^2 + \frac{t^3}{3}$: matches through t^2, but the t^3 coefficient ($\frac{1}{3}$) is smaller than the exact coefficient (1).
- $y_3(t) = 1 + t + t^2 + t^3 + \frac{2}{3}t^4 + \frac{1}{3}t^5 + \cdots$: now matches through t^3, and the higher-order coefficients are getting closer in the sense that Picard iterates converge (locally) to the true solution under the usual Picard–Lindelöf conditions.

Furthermore, the exact solution blows up at $t = 1$, so any comparison/approximation is meaningful on intervals strictly inside $(-\infty, 1)$.

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(i) $y_1(t) = e^t \sin t, \quad y_2(t) = e^t \cos t$

$$y_1'(t) = e^t \sin t + e^t \cos t = e^t(\sin t + \cos t),$$
$$y_2'(t) = e^t \cos t - e^t \sin t = e^t(\cos t - \sin t).$$

Then

$$\begin{aligned} W(y_1, y_2)(t) &= y_1 y_2' - y_2 y_1' \\ &= (e^t \sin t)(e^t(\cos t - \sin t)) - (e^t \cos t)(e^t(\sin t + \cos t)) \\ &= e^{2t}\big(\sin t \cos t - \sin^2 t - \sin t \cos t - \cos^2 t\big) \\ &= -e^{2t}\big(\sin^2 t + \cos^2 t\big) = -e^{2t}. \end{aligned}$$

(ii) $y_1(\theta) = \cos^2 \theta, \quad y_2(\theta) = 1 + \cos 2\theta$

From the trigonometric identity $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$, it follows that

$$y_2(\theta) = 1 + \cos 2\theta = 2\cos^2 \theta = 2y_1(\theta).$$

Since the two functions are linearly dependent, their Wronskian is identically zero:

$$W(\cos^2 \theta,\ 1 + \cos 2\theta)(\theta) = 0.$$

7

(i) $y'' + (\cos t)y' + 3(\ln|t|)y = 0, \quad y(2) = 3,\ y'(2) = 1$

Already in standard form:

- $p(t) = \cos t$ is continuous for all $t \in \mathbb{R}$.
- $q(t) = 3\ln|t|$ is continuous on $(-\infty, 0)$ and on $(0, \infty)$, but **not** at $t = 0$.

Since $t_0 = 2 > 0$, the **largest open interval containing** 2 that avoids the discontinuity at 0 is

$$\boxed{(0, \infty)}.$$

(ii) $(x - 3)y'' + xy' + (\ln|x|)y = 0, \quad y(1) = 0,\ y'(1) = 1$

Convert to standard form by dividing by $x - 3$ (valid only when $x \neq 3$):

$$y'' + \frac{x}{x-3}y' + \frac{\ln|x|}{x-3}y = 0.$$

Potential discontinuities:

- $\frac{x}{x-3}$ and $\frac{\ln|x|}{x-3}$ blow up at $x = 3$.
- $\ln|x|$ is not continuous at $x = 0$.

The coefficients are continuous only on intervals avoiding 0 and 3. Since $x_0 = 1$ lies between 0 and 3, the **largest open interval containing** 1 is

$$\boxed{(0, 3)}.$$

(iii) $(x - 2)y'' + y' + (x - 2)(\tan x)y = 0, \quad y(3) = 1,\ y'(3) = 2$

Divide by $x - 2$ (so $x \neq 2$):

$$y'' + \frac{1}{x-2}y' + (\tan x)y = 0.$$

Potential discontinuities:

- $\frac{1}{x-2}$ blows up at $x = 2$.
- $\tan x$ is discontinuous where $\cos x = 0$, i.e. at $x = \frac{\pi}{2} + k\pi$.

The nearest such points around $x_0 = 3$ are located:

- $x = 2$ is left of 3.
- The nearest tangent discontinuity to the right is $x = \frac{3\pi}{2} \approx 4.712$, and to the left is $x = \frac{\pi}{2} \approx 1.571$ (but 2 is closer and determines the left bound).

Thus the **largest open interval containing** 3 that avoids 2 and $\frac{3\pi}{2}$ is

$$\boxed{(2, \tfrac{3\pi}{2})}.$$

8

Use Abel's theorem:

(i) $(\cos t)y'' + (\sin t)y' - ty = 0$

Divide by $\cos t$ (on any interval where $\cos t \neq 0$):

$$y'' + \tan(t)y' - t\sec(t)y = 0 \quad \Rightarrow \quad p(t) = \tan t.$$

Then

$$W(t) = C\exp\left(-\int \tan t dt\right) = C\exp\left(-(-\ln|\cos t|)\right) = C|\cos t|.$$

On any interval where $\cos t$ keeps a fixed sign, absorb the sign into C, so

$$\boxed{W(t) = C\cos t, \qquad (\cos t \neq 0).}$$

(ii) $x^2y'' + xy' + (x^2 - \nu^2)y = 0$ **(Bessel)**

Divide by x^2 (on any interval where $x \neq 0$):

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0 \quad \Rightarrow \quad p(x) = \frac{1}{x}.$$

Hence

$$W(x) = C\exp\left(-\int \frac{1}{x}dx\right) = C\exp(-\ln|x|) = \frac{C}{|x|}.$$

On any interval not crossing 0, absorb the sign into C:

$$\boxed{W(x) = \frac{C}{x}, \qquad (x \neq 0).}$$

(iii) $(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0$ **(Legendre)**

Divide by $1 - x^2$ (on any interval where $1 - x^2 \neq 0$, i.e. $x \neq \pm 1$):

$$y'' - \frac{2x}{1 - x^2}y' + \frac{\alpha(\alpha + 1)}{1 - x^2}y = 0 \quad \Rightarrow \quad p(x) = -\frac{2x}{1 - x^2}.$$

Then

$$\int p(x)dx = \int -\frac{2x}{1 - x^2}dx = \ln|1 - x^2|,$$

so

$$W(x) = C\exp\left(-\ln|1 - x^2|\right) = \frac{C}{|1 - x^2|}.$$

On any interval avoiding ± 1, absorb sign into C:

$$\boxed{W(x) = \frac{C}{1 - x^2}, \qquad (x \neq \pm 1).}$$